



NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

School of Computer Science

Literature Review

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**You are what you do,
not what you say you'll do**

Agenda

- Getting you to think about the purpose of a literature review
- Getting you to think about what your own literature review will look like
- Critical thinking, reading and writing
- Planning and structuring options



**Review the
Available
Literature**

**Publish
Findings**

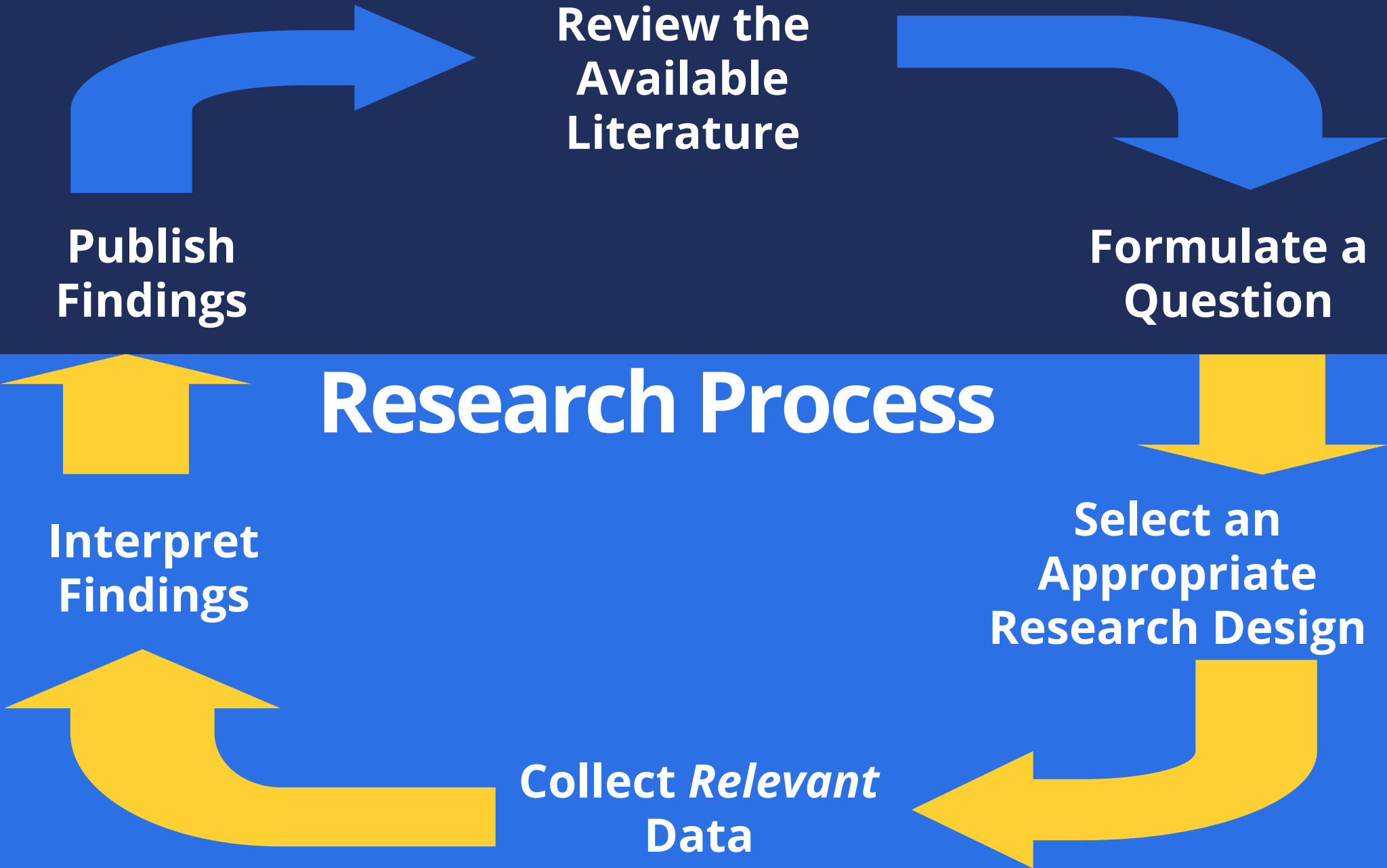
**Formulate a
Question**

Research Process

**Interpret
Findings**

**Select an
Appropriate
Research Design**

**Collect *Relevant*
Data**





What is a literature review (for)?

- When thinking about what a literature review *is*, we need to think about what it is *for*.
- What is the purpose of a literature review?
- *Activity 1:- What is literature review in your own understanding?*

What is a literature review (for)?

Establishes the terms and context.

- *How else will you define exactly what you're looking at and where its limits are?*

Presents a survey of preceding literature on the topic.

- *How else will you know what's been done already?*

Explores ways that others have solved similar questions/problems.

- *How else will you select an appropriate methodology and approach?*

Outlines the relationship of these texts to each other.

- *How else will you know what the different perspectives and debates are, and where you are coming from?*

Evaluates the quality and relevance of the literature.

- *How else will you be able to build on or reject it?*

Establishes the gaps or inadequacies.

- *How else will you justify your own contribution?*

Demonstrates your scholarly rigor.

- *How else can I have faith in your conclusions?*

Literature Review

A critical analysis of existing research in your field - highlights both the strengths and weaknesses of existing research

Allows you to gain a critical understanding of your field

Opportunity to think about what has been done in your field; opportunity to think about the similarities, patterns, trends and also differences across the existing research

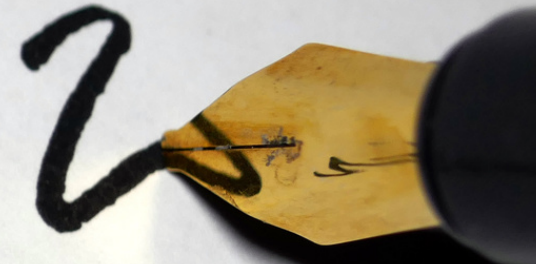
By identifying strengths and weakness, you will be able to think about what has not/needs to be done in your field

The gap in the literature is your justification for your research

MORE THAN JUST A CHAPTER...



- A literature review is a process as well as an outcome!
- **Literature review as an outcome:** appears in the final draft of your thesis as part of your introduction or as a separate chapter.
- **Literature review as a process:** critical engagement (thinking, reading and writing) with relevant research on your topic. It is a crucial and formative stage of your thesis journey.





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CRITICAL THINKING LEVELS



Undergraduate level:

backing your points up with appropriate, authoritative sources



Masters Level:

demonstrating that you understand how knowledge is created



PhD Level:

demonstrating that you can create new, original knowledge



Professional academic:

peer review, building a coherent body of work of your own and assuring others' work





READING CRITICALLY

THINK ABOUT:

What were the research aims of the paper/book?

1

Is the research aim achieved?
If so, how did they do it?

2

Are there any problems with
their methodology?

3

Was it a strong or a weak
research model?

4

How will this research help with
your own research?

5

What can you take from it?
What needs to be avoided?
What are you doing differently?



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Critical thinking

(a three-year-old's guide)

Why?

How does that
work?

What's that
made of?

But X says...

How do you
know?

What's that
for?

What does
that mean?

Says who?

What happens
if...

So what?



What critical thinking can mean in terms of the literature review

- Understanding research on its own terms – testing its viability
- Understanding research in relation to other arguments
- Critiquing research in relation to what you want to do



Starting to think about your own literature review

Who are the key players in my field? This could be anything from academics, governing bodies, schools of thought etc. (Sources!)

What are the main ideas/debates in my field?

How have these ideas changed over time?

What are some of the problems with these ideas/debates?
Is there a problem with the methodology?

What are you going to do differently?



Key players and sources

First stage of the literature review is to identify the key people in your field and collate all relevant sources about your topic.

Ask yourself:

- What research and theory is there on my topic?
- What are the key sources (books, articles) on my topic?
- Who are the main theorists and researchers in this area?
- How has the topic/problem been investigated over time?



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Where to start?!



YOU DON'T NEED TO READ
EVERYTHING – you can't!

- You don't need to read every text
- You don't need to read every word



(" Artificial Intelligence" OR "AI"
OR " Machine Learning" OR
"ML") AND (" Ethics" OR "
Governance")



Main ideas/debates

Once you have the relevant sources you can begin to think about what the key ideas, debates, methodologies etc. are in your field.

You can also think about how these ideas have changed over time.

Ask yourself:

- How has the topic or problem been defined?
- Are there any trends and patterns across the literature?
- What methodological assumptions and approaches have been used?
- What are the agreements and disagreements between theorists on my topic?



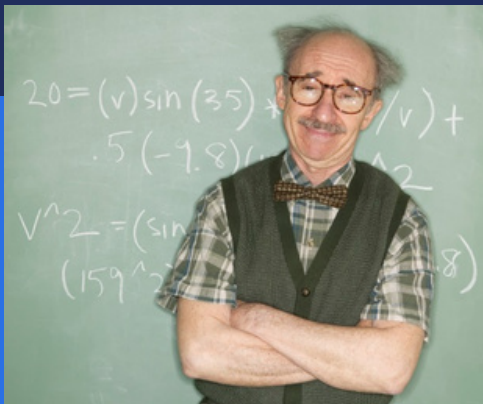
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Organising your material: Identifying a debate

Scholar X



Scholar Y



- disagrees with
- agrees with (*school of thought?*)
- builds on the conclusions of
- confirms the findings of
- has reservations about





Thinking critically

When identifying the key ideas, themes and methodologies in your field, it is important to think critically about them

This will allow you to identify a 'gap' in the literature

Ask yourself:

- What are the strengths and weaknesses of these debates?
- What evidence is lacking, inconclusive or limited?
- What will you add to the topic? What will you do differently?



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- What's your story?



Map your story (literature review):

- What is your topic?
- Who are the key people in your field?
- What are the key resources?
- What are the key ideas in your field?
- What methodologies have been used?
- What are some of the strengths and weaknesses of existing research?
- What will your contribution be?
- How will it be different?

(NB: If you can't answer some of these questions, make a note of this. It will come in handy later!)



Planning is about organising the structure of your literature review (your story will help with this!)

How will you organise the information?

- Chronologically?
- Thematically?
- By trends/approaches/techniques?
- Major debates/controversies?
- Probably a combination of these



You can:

- Jot down ideas
- Move them around
- Create links
- Put text on nodes
- Swap between 'Diagram' and 'Outline'
- Transfer to Word



Writing the review

Start with an overview

Decide on organising principles (themes, trends, methodology, chronology, controversies – usually a combination of some of these)

Use headings for the different sections of the review

Provide summative signposts of where your argument is leading

Summarise your review/highlight 'gap' in research



Tips for clear writing:

- Clear introduction: overview of topic, aim of review and structure
- Clear paragraph structure
- Make sure the subject of your sentence is clear
- Don't assume knowledge
- Make sure key terminology and difficult ideas are always explained thoroughly (ask your yourself: does it make sense?)
- Be objective and balanced
- Use signposts to orientate the reader



Where appropriate, begin sections and paragraphs with a statement which synthesises or analyses, rather than just describes

Use signposting words to demonstrate how texts relate to each other and also what you think of them

- Eg. However, yet, moreover, indeed, similarly etc



Paragraphs and flow

Paragraph:

- Topic sentence
- Discussion of topic
- Closing sentence

Thematic and grammatical links

- Logical progression from one paragraph to the next
- Demonstrate links in your language



Use of citations in the literature review

Two types of citations:

Integral: The author's name appears in the sentence.

- Example (author-date system): Lillis (2001) argues that both tutors and students often lack explicit knowledge of the conventions governing the construction of academic texts.

Non-integral: The author's name appears outside sentence.

- Example: Both tutors and students often lack explicit knowledge of the conventions governing the construction of academic texts (Lillis, 2001).



- **Whose voice is dominant - the writer's or the original author's?**
- The moon is made of cheese (Brie 1999).
- Brie (1999) argues that the moon is made of cheese.
- As Brie (1999) points out, the moon is made of cheese.
- According to Brie (1999), the moon is made of cheese.
However,
- Brie (1999) argues out that the moon is made of cheese.
However,



Learning from models

Look at samples of good theses in your field

Read reviews in prestigious journals in your field



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The End of Session

All the best in your research!!